



Lumbar Spinal Stenosis

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Lumbar spinal stenosis (LSS) is narrowing of the lumbar spinal canal causing compression of the nerve rootlets and nerve roots in the cauda equina before their exit from the foramina. It causes positional back pain, symptoms of nerve root compression in the foramina, and lower-extremity pain during walking or weight bearing.

(See also [Evaluation of Neck and Back Pain](#).)

Spinal stenosis can be congenital or acquired. It may involve the cervical or lumbar spine. Lumbar spinal stenosis often occurs through increases in the size of the ligamentum flavum, enlargement of the facet joints, and broad-based disc bulges. The most common causes of LSS are [osteoarthritis](#), degenerative disc disorders, [spondylosis](#), and [spondylolisthesis](#) with compression of the cauda equina. Other causes include [Paget disease of bone](#)

Symptoms and Signs of Lumbar Spinal Stenosis

In patients with lumbar spinal stenosis, pain occurs in the buttocks, thighs, or calves during walking, running, climbing stairs, or even standing. This pain is referred to as neurogenic claudication. The pain is not relieved by standing still but by flexing the back or by sitting (although paresthesias may continue) because, as with the cervical spinal canal, the lumbar spinal canal is larger in flexion and smaller in extension. Thus, patients often experience some relief when pushing a shopping cart or using a rolling walker because the back is slightly flexed. Similarly, walking up hills is less painful than walking down. Patients may have pain, paresthesias, weakness, and diminished reflexes in the affected nerve root distribution.

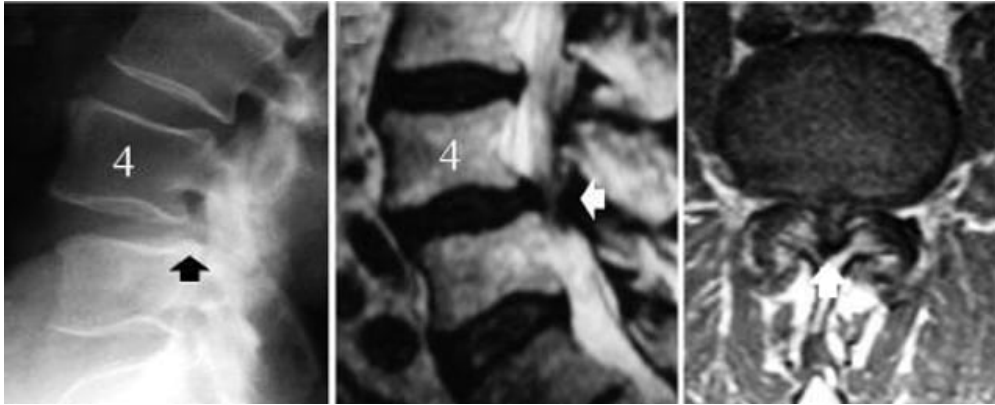
Rarely, sudden nerve rootlet compression resulting from LSS or a large disc herniation may cause [cauda equina syndrome](#) with distal leg paresis and sensory loss in and around the perineum and anus (saddle anesthesia), as well as bladder, bowel, and pudendal dysfunction; unlike in spinal cord injury, muscle tone and deep tendon reflexes are decreased in the legs.

Diagnosis of Lumbar Spinal Stenosis

- Clinical evaluation
- Sometimes MRI, electrodiagnostic studies, or both

Lumbar Spinal Stenosis Caused by Degenerative Spondylolisthesis of L...

IMAGE



The image on the left is a lateral radiograph of the lumbar spine showing slipping of the L4 vertebra (black arrow—the posterior border of the L5 vertebral body does not align with that of L4). The image in the middle is a midsagittal MRI scan showing compression of the neural structures (white arrow). The image on the right is an axial MRI scan showing severe spinal stenosis at the level of the degenerative spondylolisthesis, L4-5 (white arrow).

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Spinal stenosis is suspected based on characteristic symptoms of pain that is increased by exertion (neurogenic claudication) and relieved with sitting and/or with back flexion. Muscle strength is usually maintained, but focal weakness, sensory loss, and diminished deep tendon reflexes may be present.

Calf symptoms may simulate those of intermittent vascular claudication. Vascular claudication can be differentiated by relief with rest (not position change), skin atrophy, and abnormalities in pulses, capillary refill, and vascular tests.

Diagnostic tests for lumbar spinal stenosis, when needed, are [the same as for lumbosacral radiculopathy](#), including MRI or CT and electrodiagnostic studies.

Treatment of Lumbar Spinal Stenosis

- Activity as tolerated, analgesics, and sometimes medications that relieve neuropathic pain
- Physical therapy
- Possibly epidural corticosteroid injections
- Surgery for severe cases

In patients with lumbar spinal stenosis, conservative treatments and indications for surgery are similar to those for [lumbosacral radiculopathy](#).

Epidural corticosteroid injections sometimes provide transient relief. In symptomatic patients who are poor candidates for surgical intervention, the combination of epidural injections and physical therapy can result in some symptomatic improvement.

For advanced spinal stenosis, surgery involves decompression of nerve root entrapment by vertebral canal and foraminal encroachments, which sometimes requires laminectomy at 2 or 3 levels plus foraminotomies and sometimes fusion surgery.

Spinal stability must be preserved. Spinal fusion may be indicated if there is instability or if there are severe, well-localized arthritic changes in 1 or 2 vertebral interspaces; however, some studies highlight the controversial nature of this approach ([1](#), [2](#)).

Treatment references

1. [Försth P, Olafsson G, Carlsson T, et al](#). A randomized, controlled trial of fusion surgery for lumbar spinal stenosis. *N Engl J Med*. 374:1413-1423, 2016. doi: 10.1056/NEJMoa1513721
2. [Ghogawala Z, Dziura J, Butler WE, et al](#). Laminectomy plus fusion versus laminectomy alone for lumbar spondylolisthesis. *N Engl J Med*. 374:1424-1434, 2016. doi: 10.1056/NEJMoa1508788

Key Points

- Lumbar spinal stenosis often occurs through increases in the size of the ligamentum flavum, enlargement of the facet joints, and broad-based disc bulges.
- The most common causes are [osteoarthritis](#), degenerative disc disorders, [spondylosis](#), and [spondylolisthesis](#) with compression of the cauda equina.
- Symptoms include pain in the buttocks, thighs, or calves that worsens with standing and walking and is relieved by bending or leaning forward, which is referred to as neurogenic claudication.
- Diagnose based on characteristic symptoms along with MRI, electrodiagnostic studies, or both.
- Recommend analgesics, light activity as tolerated, and physical therapy; however, if pain or deficits are severe or worsening, consider surgical procedures.